

Review of Last Lecture

7.1 Definition of z-Transform

7.2 z-Transform of Typical Sequences

7.3 Region of Convergence of z-Transform

7.4 Inverse z-Transform

7.1 Definition of z-Transform

For a continuous causal signal $x(t)$ sampled at interval T , the sampled signal $x_s(t)$ is

$$x_s(t) = x(t)\delta_T(t) = \sum_{n=0}^{\infty} x(nT)\delta(t - nT)$$

Unilateral Laplace Transform

$$X_s(s) = \int_0^{\infty} x_s(t)e^{-st}dt = \sum_{n=0}^{\infty} x(nT) \int_0^{\infty} \delta(t - nT)e^{-st}dt = \sum_{n=0}^{\infty} x(nT)e^{-snT}$$

Let $z = e^{sT}$, where **z is a complex variable**. The relationship between the Laplace transform of the sampled signal and the z-transform of the sequence is as follows:

$$X(z) \Big|_{z=e^{sT}} = X_s(s)$$

Usually let $T=1$, $z = e^s$

**Unilateral
z-Transform**

$$X(z) = \sum_{n=0}^{\infty} x(n)z^{-n}$$

**Bilateral
z-Transform**

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

The z-transform of a sequence is **a power series of the complex variable z^{-1}** , and **its coefficients are $x(n)$** .

Which of the following statements are correct? ()

A

The z-transform is helpful to solve difference equations.

B

For a causal sequence, the unilateral z-transform is equivalent to the bilateral z-transform.

C

The unilateral z-transform of an anti-causal sequence is equivalent to the bilateral z-transform.

D

The unilateral z-transform of an anti-causal sequence is zero.

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7.2 z-Transform of Typical Sequences

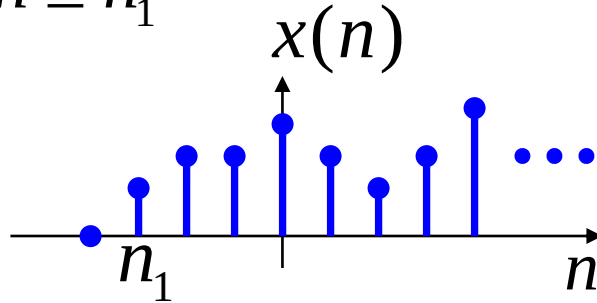
Sequence	Z-Transform	Type of Z-Transform
$\delta(n)$	1 (Converges over the entire z-plane)	Unilateral/Bilateral Z-Transform
$\delta(n-1)$	$z^{-1} \quad (z > 0)$	Unilateral/Bilateral Z-Transform
$\delta(n+1)$	0	Unilateral Z-Transform
	$z \quad (z < \infty)$	Bilateral Z-Transform
$u(n)$	$\frac{z}{z-1} \quad (z > 1)$	Unilateral/Bilateral Z-Transform
$a^n u(n)$	$\frac{z}{z-a} \quad (z > a)$	Unilateral/Bilateral Z-Transform
$nu(n)$	$\frac{z}{(z-1)^2} \quad (z > 1)$	Unilateral/Bilateral Z-Transform

7.2 z-Transform of Typical Sequences

Sequence	Z-Transform	Type of Z-Transform
$e^{-j\omega_0 n} u(n)$	$\frac{z}{z - e^{-j\omega_0}} \quad (z > 1)$	Unilateral/Bilateral Z-Transform
$\beta e^{-j\omega_0 n} u(n)$	$\frac{z}{z - \beta e^{-j\omega_0}} \quad (z > \beta)$	Unilateral/Bilateral Z-Transform
$\cos(\omega_0 n) u(n)$	$\frac{z(z - \cos \omega_0)}{z^2 - 2z \cos \omega_0 + 1} \quad (z > 1)$	Unilateral/Bilateral Z-Transform
$\sin(\omega_0 n) u(n)$	$\frac{z \sin \omega_0}{z^2 - 2z \cos \omega_0 + 1} \quad (z > 1)$	Unilateral/Bilateral Z-Transform
$\beta^n \cos(\omega_0 n) u(n)$	$\frac{z(z - \beta \cos \omega_0)}{z^2 - 2z \beta \cos \omega_0 + \beta^2} \quad (z > \beta)$	Unilateral/Bilateral Z-Transform
$\beta^n \sin(\omega_0 n) u(n)$	$\frac{z \beta \sin \omega_0}{z^2 - 2z \beta \cos \omega_0 + \beta^2} \quad (z > \beta)$	Unilateral/Bilateral Z-Transform

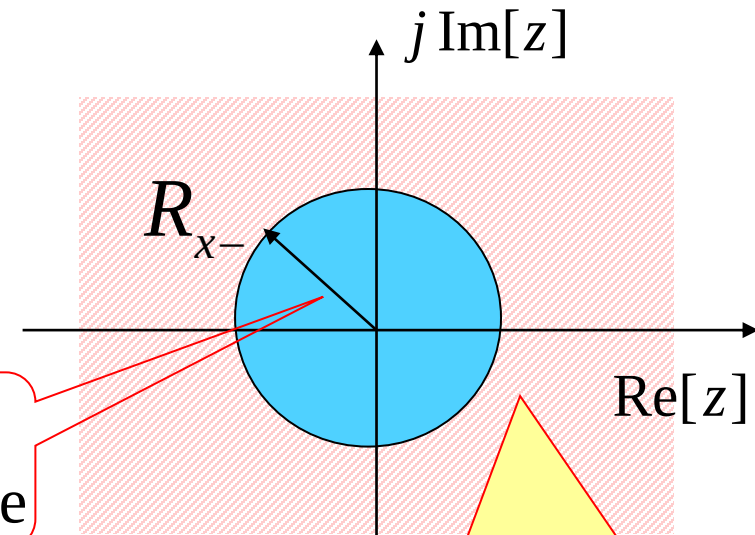
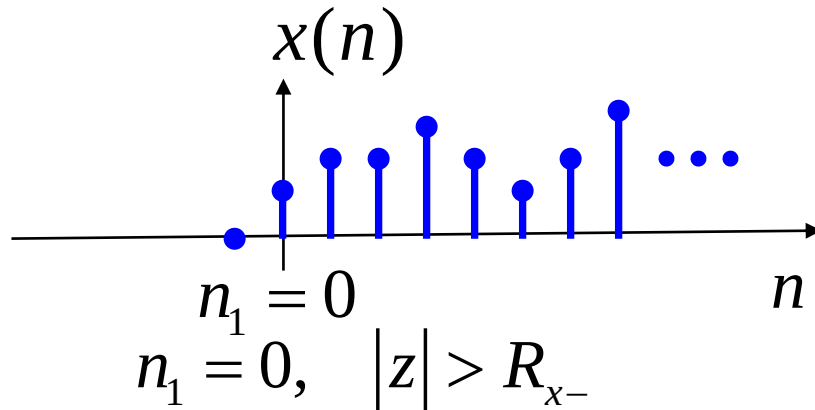
7.3 Region of Convergence of z-Transform

1. Right-Sided Sequence: A sequence that has non-zero finite values only in the interval $n \geq n_1$



$$X(z) = \sum_{n=n_1}^{\infty} x(n)z^{-n} \quad n_1 \leq n \leq \infty$$

$n_1 < 0$, $R_{x-} < |z| < \infty$ (The ROC does not include $|z| = \infty$, because ∞^{-n_1} is infinite)

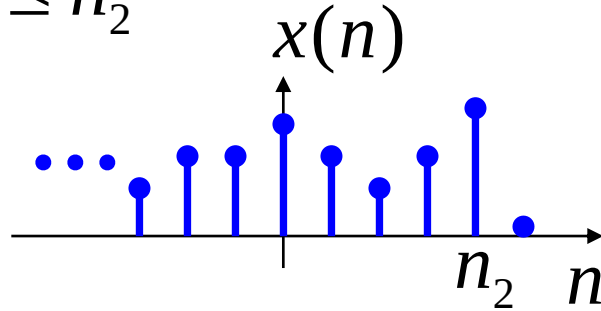


The ROC is outside the circle. If $n_1 < 0$, $|z| = \infty$ is not included.

A **causal sequence** is a special case of a right-sided sequence, and its ROC is $|z| > R_{x-}$

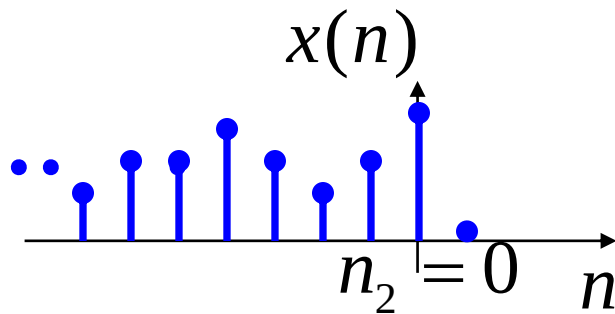
7.3 Region of Convergence of z-Transform

2. Left-Sided Sequence: A sequence $x(n)$ that has non-zero finite values only in the interval $n \leq n_2$



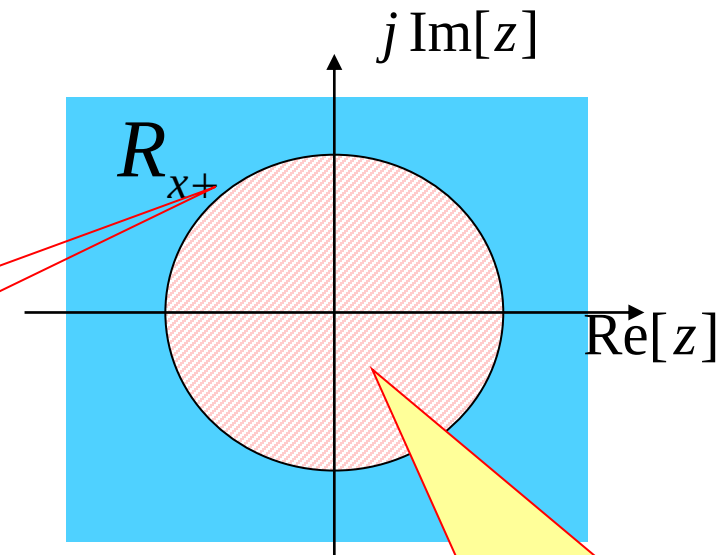
$$X(z) = \sum_{n=-\infty}^{n_2} x(n)z^{-n} \quad -\infty \leq n \leq n_2$$

$n_2 > 0$, $0 < |z| < R_{x+}$ (The ROC does not include $z = 0$, because the negative power of 0 is meaningless)



$$n_2 = 0, \quad |z| < R_{x+}$$

Radius of Convergence

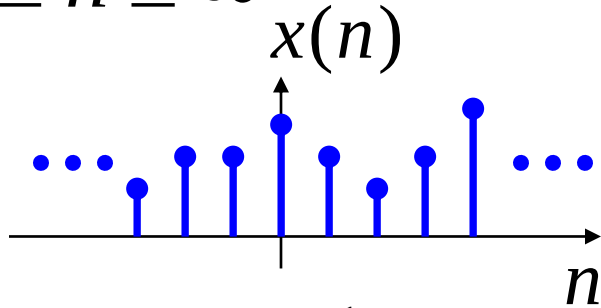


The ROC is inside the circle. If $n_2 > 0$, $z=0$ is not included.

7.3 Region of Convergence of z-Transform

3. Double-Sided Sequence: A sequence that has non-zero finite values in the interval

$$-\infty \leq n \leq \infty$$



$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n} \quad -\infty \leq n \leq \infty$$

$$X(z) = \sum_{n=-\infty}^{-1} x(n)z^{-n} + \sum_{n=0}^{\infty} x(n)z^{-n}$$

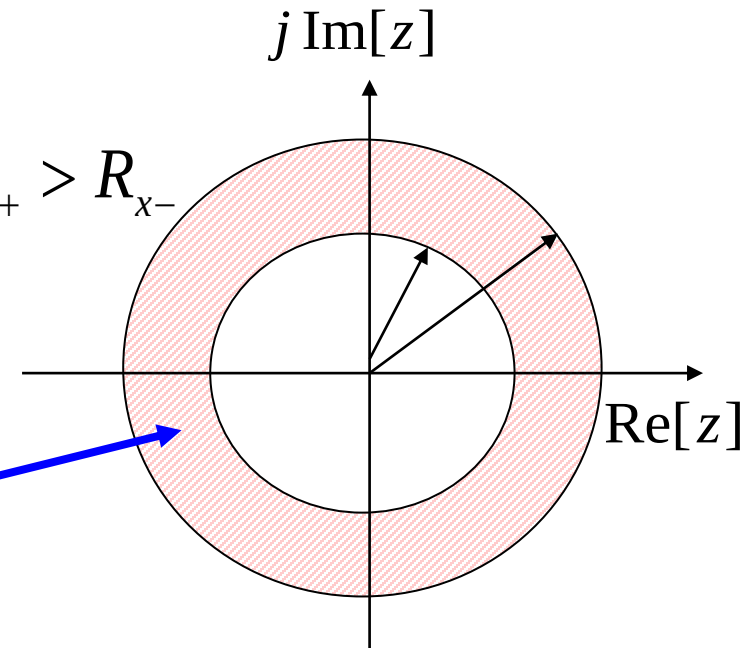
Converges inside
the circle
 $|z| < R_{x+}$

Converges outside
the circle
 $|z| > R_{x-}$

$$R_{x+} > R_{x-}$$

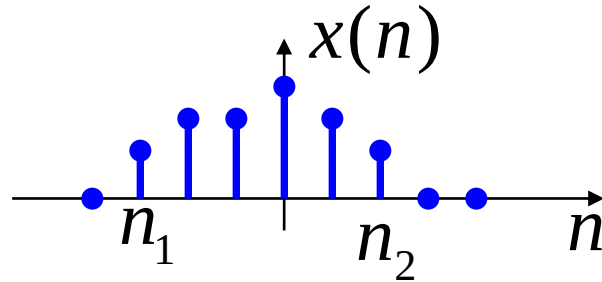
$R_{x+} > R_{x-}$ There is an annular ROC

$R_{x+} < R_{x-}$ No ROC



7.3 Region of Convergence of z-Transform

4. Finite-Length Sequence: A sequence that has non-zero finite values in the finite interval $n_1 \leq n \leq n_2$



$$X(z) = \sum_{n=n_1}^{n_2} x(n)z^{-n} \quad n_1 \leq n \leq n_2$$

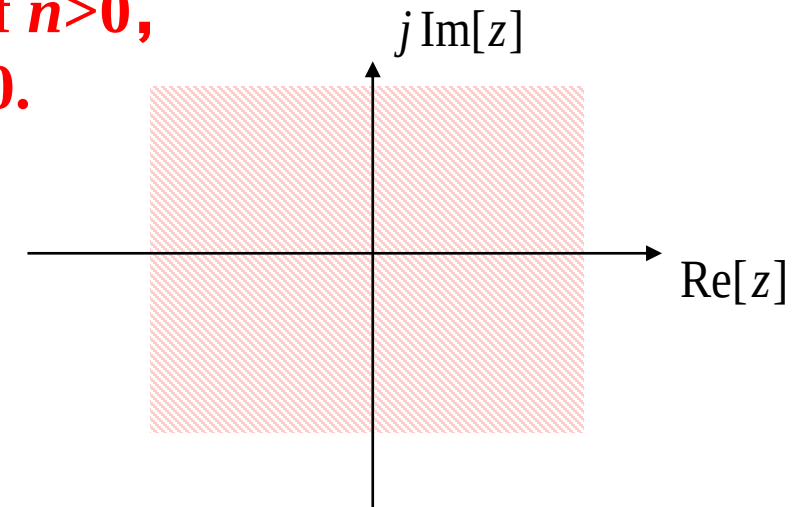
Divided into three cases:

$$n_1 < 0, n_2 > 0 \quad 0 < |z| < \infty$$

$$n_1 \geq 0, n_2 > 0 \quad 0 < |z| \leq \infty$$

$$n_1 < 0, n_2 \leq 0 \quad 0 \leq |z| < \infty$$

**0^{-n} is meaningless if $n > 0$,
 ∞^{-n} is infinite if $n < 0$.**



The ROC is at least the entire z-plane but 0 and ∞ (If pole-zero cancellation occurs, the ROC expands to the entire z-plane).

Which of the following statements are correct? ()

A

Using the partial fraction method to find the inverse transform of $X(z)$ requires that $X(z)$ is a proper fraction.

B

Sequence shifting does not affect the zeros and poles of the z-transform (except at 0 and ∞)

C

The condition for applying the final value theorem is that all poles must lie inside the unit circle.

D

When pole-zero cancellation occurs, the ROC of $X(z)$ may include poles.

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7.4 Inverse z-Transform

$$X(z) = ZT[x(n)] = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

Partial Fraction Expansion Method:

$$X(z) = \frac{b_0 + b_1z + \dots + b_{r-1}z^{r-1} + b_rz^r}{a_0 + a_1z + \dots + a_{k-1}z^{k-1} + a_kz^k}$$

1. $X(z)$ has only first-order poles

$$X(z) = A_0 + \sum_{m=1}^M \frac{A_m z}{z - z_m} = A_0 + \frac{A_1 z}{z - z_1} + \dots + \frac{A_M z}{z - z_M} \quad A_0 = X(z) \Big|_{z=0} = \frac{b_0}{a_0}$$

Let z_m be a first-order pole of $X(z)$ $A_m = \left[(z - z_m) \frac{X(z)}{z} \right]_{z=z_m}$ $X(z) = A_0 + \sum_{m=1}^M \frac{z A_m}{z - z_m}$

or $X(z) = A_0 + \sum_{m=1}^M \frac{A_m}{1 - z_m z^{-1}} = A_0 + \frac{A_1}{1 - z_1 z^{-1}} + \dots + \frac{A_M}{1 - z_M z^{-1}} \quad A_m = \left[(1 - z_m z^{-1}) X(z) \right]_{z=z_m}$

Inverse Transform: $x(n) = A_0 \delta(n) + \sum_{m=1}^M A_m z_m^n = A_0 \delta(n) + A_1 z_1^n + \dots + A_M z_M^n$

2. In addition to M first-order poles, $X(z)$ also has an N -th order pole at $z=z_i$

$$X(z) = A_0 + \sum_{m=1}^M \frac{A_m z}{z - z_m} + \sum_{j=1}^N \frac{B_j z}{(z - z_i)^j} \quad (|z| > \max\{z_1, \dots, z_M, z_i\})$$

$$\text{Let } X_1(z) = (z - z_i)^N \frac{X(z)}{z}$$

$$B_N = X_1(z) \Big|_{z=z_i} \quad B_j = \frac{1}{(N-j)!} \left[\frac{d^{N-j}}{dz^{N-j}} X_1(z) \right]_{z=z_i} \quad (j = 1, \dots, N-1)$$

$$\text{If it is a second-order pole } N = 2: \quad X_1(z) = (z - z_i)^2 \frac{X(z)}{z}$$

$$B_2 = X_1(z) \Big|_{z=z_i} \quad B_1 = \left[\frac{d}{dz} X_1(z) \right]_{z=z_i}$$

Inverse transform: $x(n) = A_0 \delta(n) + (A_1 z_1^n + \dots + A_M z_M^n + B_1 z_i^n + B_2 n z_i^{n-1}) u(n)$

7.4 Inverse z-Transform

Generally, $X(z)$ can be written as:

$$X(z) = \frac{b_0 + b_1 z + \cdots + b_{r-1} z^{r-1} + b_r z^r}{a_0 + a_1 z + \cdots + a_{k-1} z^{k-1} + a_k z^k}$$

When using the **partial fraction expansion** method to find the inverse transform, you need to master the inverse transforms of basic forms.

Note: The inverse transform is related to the ROC.

$$\frac{z}{z - z_m} (|z| > |z_m|) \Leftrightarrow z_m^n u(n)$$

$$\frac{z}{z - z_m} (|z| < |z_m|) \Leftrightarrow -z_m^n u(-n-1)$$

Discussion: Only **proper fractions** can be expanded by **partial fractions**, but the expanded form needs to be **multiplied by z** to have the basic form of the inverse z-transform mentioned above.

To perform partial fraction expansion on $\frac{X(z)}{z}$, $\frac{X(z)}{z}$ is required to be a proper fraction — which means **$k \geq r$ is needed to ensure $X(z)$ converges at $z=\infty$. The ROC of the z-transform of a causal sequence is $|z| > R_x$, and $k \geq r$ is a necessary and sufficient condition for convergence.**

Contents of This Lecture

7.5 Basic Properties of the z-Transform

7.6 Relationship Between the z-Transform
and the Laplace Transform

Learning Objectives of This Lecture

1. Master the **basic properties of the z-transform** proficiently;
2. Be able to analyze the impact on the **Region of Convergence** (ROC) when applying the basic properties of the z-transform;
3. Be familiar with the mapping relationship between the **s-plane** and the **z-plane**.

7.5 Basic Properties of the z-Transform

7.5.1 Linearity

$$\text{if } ZT[x(n)] = X(z) \quad (R_{x1} < |z| < R_{x2}),$$

$$ZT[y(n)] = Y(z) \quad (R_{y1} < |z| < R_{y2})$$

$$\text{then } ZT[ax(n) + by(n)] = aX(z) + bY(z)$$

$$\max(R_{x1}, R_{y1}) < |z| < \min(R_{x2}, R_{y2})$$

the ROC is the overlapping part

Note: If **some zeros and poles cancel each other out** in the linear combination,
the ROC may expand.

7.5.2 Shift Property

1. Bilateral z-transform of a Shifted Bilateral Sequence

$$ZT[x(n)] = X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

$$ZT[x(n-m)] = z^{-m} X(z)$$

$$ZT[x(n+m)] = z^m X(z)$$

m is any positive integer.

If $x(n)$ is a double-sided sequence and its bilateral z-transform is $X(z)$, shifting the sequence $x(n)$ will change the ROC of $X(z)$. This statement is ().

☐ A True

☒ B False

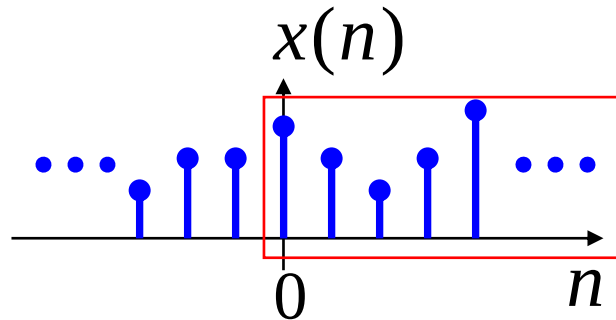
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7.5 Basic Properties of the z-Transform

2. Unilateral z-transform of a Left-Shifted Bilateral Sequence

$$X(z) = \sum_{n=0}^{\infty} x(n)u(n)z^{-n}$$

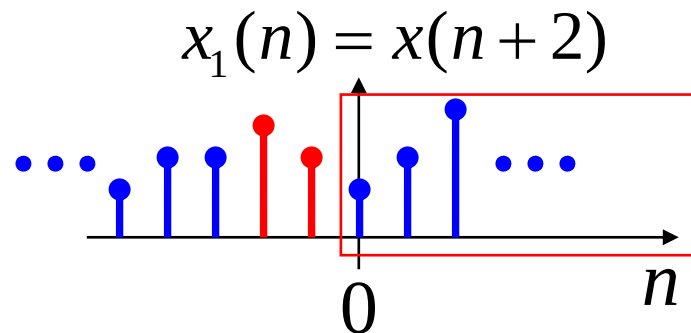
$$ZT[x(n+m)u(n)] = \sum_{n=0}^{\infty} x(n+m)z^{-n}$$



$$= z^m \sum_{n=0}^{\infty} x(n+m)z^{-(n+m)} = z^m \sum_{k=m}^{\infty} x(k)z^{-k}$$

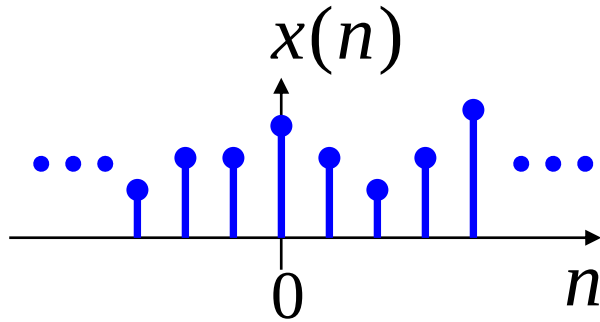
$$= z^m \left[\sum_{k=0}^{\infty} x(k)z^{-k} - \sum_{k=0}^{m-1} x(k)z^{-k} \right]$$

$$= z^m \left[X(z) - \sum_{k=0}^{m-1} x(k)z^{-k} \right]$$



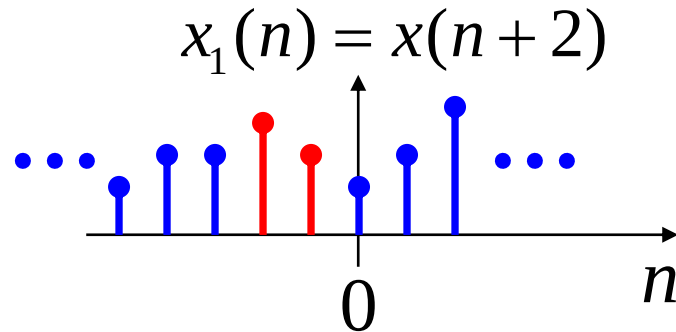
**reduces the contribution
of some points**

7.5 Basic Properties of the z-Transform



$$X(z) = x(0) + x(1)z^{-1} + x(2)z^{-2} + \dots$$

$$z^2 X(z) = x(0)z^2 + x(1)z + x(2) + x(3)z^{-1} + \dots$$



$$X_1(z) = x_1(0) + x_1(1)z^{-1} + x_1(2)z^{-2} + \dots$$

$$= x(2) + x(3)z^{-1} + x(4)z^{-2} + \dots$$

$$= z^2 X(z) - x(0)z^2 - x(1)z$$

$$= z^2 \left[X(z) - x(0) - x(1)z^{-1} \right]$$

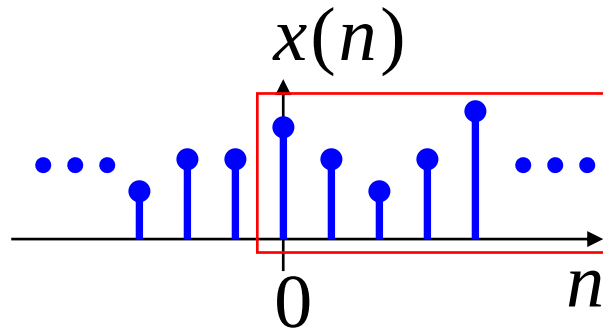
7.5 Basic Properties of the z-Transform

3. Unilateral z-Transform of a Right-Shifted Bilateral Sequence

$$X(z) = \sum_{n=0}^{\infty} x(n)u(n)z^{-n}$$

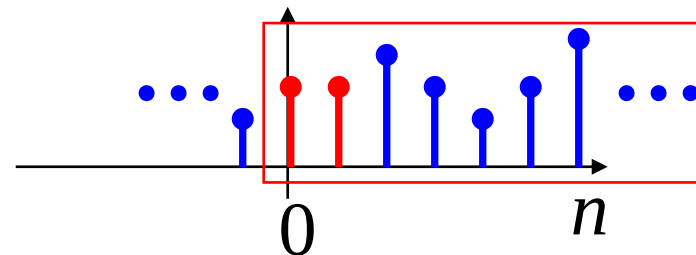
$$ZT[x(n-m)u(n)] = z^{-m} \left[X(z) + \sum_{k=-m}^{-1} x(k)z^{-k} \right]$$

adds the contribution of some points



$$X(z) = x(0) + x(1)z^{-1} + x(2)z^{-2} + \dots$$

$$x_2(n) = x(n-2)$$



$$X_2(z) = x_2(0) + x_2(1)z^{-1} + x_2(2)z^{-2} + \dots$$

$$= x(-2) + x(-1)z^{-1} + x(0)z^{-2} + \dots$$

$$= z^{-2}X(z) + x(-2) + x(-1)z^{-1}$$

$$= z^{-2} \left[X(z) + x(-2)z^2 + x(-1)z \right]$$

7.5 Basic Properties of the z-Transform

4. Unilateral z-Transform of a Shifted Causal Sequence $x(n)$

$$\because \sum_{k=-m}^{-1} x(k)z^{-k} = 0$$

$$ZT[x(n-m)u(n)] = z^{-m} X(z)$$

$$ZT[x(n+m)u(n)] = z^m \left[X(z) - \sum_{k=0}^{m-1} x(k)z^{-k} \right]$$

Sequence shifting only changes the zero-pole situation of the z-transform at **$z=0$ and ∞** .

Given the sequence $x(n) = \left(\frac{1}{3}\right)^n [u(n) - u(n - 8)]$, its ROC is ()

- ☐ A $0 < |z| < \frac{1}{3}$
- ☐ B $|z| > \frac{1}{3}$
- ☒ C $|z| > 0$
- ☐ D The entire z -plane

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7.5 Basic Properties of the z-Transform

Example 7-5: Find the z-transform and ROC of $x(n) = \left(\frac{1}{3}\right)^n [u(n) - u(n-8)]$

Solution: The sequence is a **finite-length sequence**.

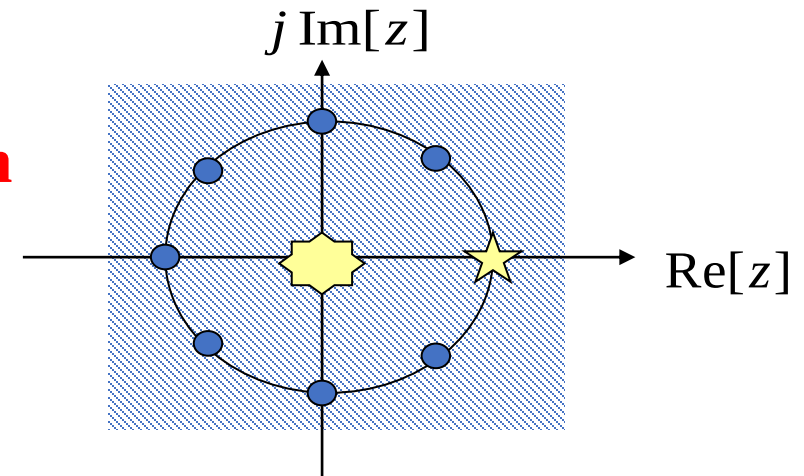
$$X(z) = \sum_{n=0}^7 \left(\frac{1}{3}\right)^n z^{-n} = \sum_{n=0}^7 \left(\frac{1}{3} z^{-1}\right)^n = \frac{1 - \left(\frac{1}{3} z^{-1}\right)^8}{1 - \frac{1}{3} z^{-1}} = \frac{z^8 - \left(\frac{1}{3}\right)^8}{z^7 \left(z - \frac{1}{3}\right)}$$

$$z^8 = \left(\frac{1}{3}\right)^8 \quad z = \frac{1}{3} e^{j\frac{2k\pi}{8}} \quad (k = 0, \dots, 7) \longrightarrow \text{Eight zeros}$$

$$z = 0 \longrightarrow \text{Seventh-order pole}$$

$$z = 1/3 \longrightarrow \text{First-order pole, which cancels with one zero (expanding the ROC beyond } |z| > 1/3)$$

$$\therefore |z| > 0$$



7.5 Basic Properties of the z-Transform

7.5.3 z-Domain Differentiation (Time-Domain Linear Weighting)

$$\text{if } ZT[x(n)] = X(z)$$

$$ZT[nx(n)] = -z \frac{d}{dz} X(z)$$

Multiplying the time-domain sequence by n is equivalent to differentiating in the z-domain and multiplying by $(-z)$.

$$ZT[n^2 x(n)] = -z \frac{d}{dz} \left[-z \frac{dX(z)}{dz} \right] = \left[-z \frac{d}{dz} \right]^2 X(z)$$

$$ZT[n^m x(n)] = \left[-z \frac{d}{dz} \right]^m X(z)$$

7.5 Basic Properties of the z-Transform

Example 7–6: Find the z-transforms of the ramp sequence $nu(n)$ and the sequence $n^2u(n)$.

Solution: $u(n) \xrightarrow{z} \frac{z}{z-1} \quad (|z| > 1)$

$$nu(n) \xrightarrow{z} -z \frac{d}{dz} \left(\frac{z}{z-1} \right) = \frac{z}{(z-1)^2} \quad (|z| > 1)$$

$$\begin{aligned} n^2u(n) \xrightarrow{z} \left[-z \frac{d}{dz} \right]^2 \left(\frac{z}{z-1} \right) &= -z \frac{d}{dz} \left[\frac{z}{(z-1)^2} \right] \\ &= \frac{z(z+1)}{(z-1)^3} \quad (|z| > 1) \end{aligned}$$

7.5 Basic Properties of the z-Transform

Common Sequence z-Transforms Obtained Using the z-Domain Differentiation Property

Z-Transform	Causal Sequence	Z-Transform	Causal Sequence
$\frac{z}{z-1} \quad (z > 1)$	$u(n)$	$\frac{z}{(z-1)^2} \quad (z > 1)$	$nu(n)$
$\frac{z}{z-a} \quad (z > a)$	$a^n u(n)$	$\frac{az}{(z-a)^2} \quad (z > a)$	$na^n u(n)$
$\frac{z^2}{(z-a)^2} \quad (z > a)$	$(n+1)a^n u(n)$	$\frac{a^2 z}{(z-a)^3} \quad (z > a)$	$\frac{n(n-1)}{2!} a^n u(n)$
$\frac{z^3}{(z-a)^3} \quad (z > a)$	$\frac{(n+1)(n+2)}{2!} a^n u(n)$		

Right-shift the sequence by 1
 multiply by coefficient a

Right-shift the sequence by 2
 multiply by coefficient a^2

Note: Right – shift $(n+1)a^n u(n)$ by 1, multiply by a , resulting in $na^n u(n-1) = 0 \cdot a^0 \delta(n) + 1 \cdot a \delta(n-1) + 2a^2 \delta(n-2) + \dots = na^n u(n)$

If the ROC of the z-transform is $|z| < |a|$, the inverse transform is an anticausal sequence: add a **negative sign** before the sequences in the above table, and replace $u(n)$ with $u(-n-1)$.

7.5 Basic Properties of the z-Transform

7.5.4 z-Domain Scaling (Time-Domain Exponential Weighting)

$$ZT[x(n)] = X(z) \quad (R_{x1} < |z| < R_{x2})$$

$$ZT[a^n x(n)] = X\left(\frac{z}{a}\right) \quad (R_{x1} < \left|\frac{z}{a}\right| < R_{x2} \Rightarrow |a|R_{x1} < |z| < |a|R_{x2}) \quad |a| > 1$$

$$ZT[a^{-n} x(n)] = X(az) \quad (R_{x1} < |az| < R_{x2} \Rightarrow \frac{R_{x1}}{|a|} < |z| < \frac{R_{x2}}{|a|}) \quad |a| > 1$$

$$ZT[(-1)^n x(n)] = X(-z) \quad (R_{x1} < |z| < R_{x2})$$

$$ZT[(-1)^n u(n)] = z/(z+1) \quad (|z| > 1)$$

Example 7-7: Find the z – transform of the sequence $\beta^n \cos(n\omega_0)u(n)$.

Solution: $\cos(n\omega_0)u(n) \xrightarrow{z} \frac{z(z - \cos \omega_0)}{z^2 - 2z \cos \omega_0 + 1}$

then

$$\beta^n \cos(\omega_0 n)u(n) \xrightarrow{z} \frac{\frac{z}{\beta} \left(\frac{z}{\beta} - \cos \omega_0 \right)}{\left(\frac{z}{\beta} \right)^2 - 2 \frac{z}{\beta} \cos \omega_0 + 1}$$
$$\rightarrow \frac{z(z - \beta \cos \omega_0)}{z^2 - 2z\beta \cos \omega_0 + \beta^2}$$

$$\left| \frac{z}{\beta} \right| > 1 \quad |z| > |\beta|$$

7.5 Basic Properties of the z-Transform

7.5.5 Initial Value Theorem

If $x(n)$ is a causal sequence, and $ZT[x(n)] = X(z) = \sum_{n=0}^{\infty} x(n)z^{-n}$

then $x(0) = \lim_{z \rightarrow \infty} X(z)$ (the signal jump at $n=0$ corresponds to high-frequency components)

7.5.6 Final Value Theorem

If $x(n)$ is a causal sequence, and $ZT[x(n)] = X(z) = \sum_{n=0}^{\infty} x(n)z^{-n}$

then $\lim_{n \rightarrow \infty} x(n) = \lim_{z \rightarrow 1} [(z-1)X(z)]$ ($n \rightarrow \infty$ corresponds to the DC component, i.e., $s=0$, $z=e^s=1$)

It requires that $x(n)$ converges as $n \rightarrow \infty$, i.e., the poles of $X(z)$ must lie **inside the unit circle (if on the unit circle, it should be a first-order pole and can only be at $z=1$)**.

Note the difference from the system stability condition: the condition **for the stability of a causal system** is that the poles of the system function must **lie inside the unit circle**.

7.5 Basic Properties of the z-Transform

The condition for using **the Final Value Theorem** is that the ROC includes the unit circle or has a first-order pole at $z=1$.

$x(n)$	Final Value ($n \rightarrow \infty$)	$X(z)$	ROC
$(2)^n$	No final value	$\frac{z}{z-2}$	$ z > 2$
$(1)^n$	Final value = 1	$\frac{z}{z-1}$	$ z > 1$
$(-1)^n$	No final value	$\frac{z}{z+1}$	$ z > 1$
$(0.5)^n$	Final value = 0	$\frac{z}{z-0.5}$	$ z > 0.5$

7.5 Basic Properties of the z-Transform

Example 7-8: Given the z-transform of a causal sequence $X(z) = \frac{1}{1-az^{-1}}$, $|z| > |a|$ (where a is a real number), find the initial value $x(0)$ and $x(\infty)$ of the sequence using the Initial Value Theorem and Final Value Theorem respectively, and derive $x(1)$.

Solution: From the Initial Value Theorem and Final Value Theorem:

$$x(0) = \lim_{z \rightarrow \infty} X(z) = \lim_{z \rightarrow \infty} \frac{1}{1-az^{-1}} = 1$$

$$x(\infty) = \lim_{z \rightarrow 1} (z-1)X(z) = \lim_{z \rightarrow 1} \frac{z-1}{1-az^{-1}} = \lim_{z \rightarrow 1} \frac{z(z-1)}{z-a} = \begin{cases} 1, & a = 1 \\ 0, & |a| < 1 \end{cases}$$

The final value does not exist when $|a| > 1$ or $a = -1$.

Since $x(1) = x(n+1)|_{n=0}$ and $x(n+1) \leftrightarrow z[X(z) - x(0)]$ (shift property of the unilateral z-transform)

$$\text{so } x(1) = \lim_{z \rightarrow \infty} z[X(z) - x(0)] = \lim_{z \rightarrow \infty} \left(\frac{z}{1-az^{-1}} - z \right) = \lim_{z \rightarrow \infty} \left(\frac{a}{1-az^{-1}} \right) = a$$

7.5 Basic Properties of the z-Transform

7.5.7 Time-Domain Convolution Theorem

Given two sequences $x(n)$ and $h(n)$, their z-transforms are:

$$ZT[x(n)] = X(z) \quad (R_{x1} < |z| < R_{x2})$$

$$ZT[h(n)] = H(z) \quad (R_{h1} < |z| < R_{h2})$$

Then the z-transform of the time-domain convolution of the two sequences is equivalent to the product of their z-transforms in the z-domain:

$$ZT[x(n) * h(n)] = X(z)H(z)$$
$$\max(R_{x1}, R_{h1}) < |z| < \min(R_{x2}, R_{h2})$$

The ROC is the overlapping part.

Note: If some zeros and poles cancel each other out, the ROC may expand.

7.5 Basic Properties of the z-Transform



$$\begin{aligned}\text{Proof: } ZT[x(n) * h(n)] &= ZT\left[\sum_{m=-\infty}^{\infty} x(m)h(n-m)\right] \\ &= \sum_{m=-\infty}^{\infty} x(m)ZT[h(n-m)] \\ &= \sum_{m=-\infty}^{\infty} x(m)z^{-m}H(z) \\ &= X(z)H(z)\end{aligned}$$

$$\begin{aligned}\text{Example: } ZT\left[\sum_{k=0}^n x(k)\right] &= ZT\{[x(n)u(n)] * u(n)\} \\ &= \frac{z}{z-1} X(z)\end{aligned}$$

Given that the excitation of a discrete-time system is $x(n) = u(n)$, and the unit sample response is $h(n) = a^n u(n) - a^{n-1} u(n-1)$ ($|a| < 1$), the ROC of the z-transform of the zero-state response is ()

- ☐ A $|a| < |z| < 1$
- ☐ B $|z| < |a|$
- ☒ C $|z| > |a|$
- ☐ D $|z| > 1$

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7.5 Basic Properties of the z-Transform

Example 7-9: Given that the excitation of a discrete-time system is $x(n) = u(n)$, and the unit sample response is $h(n) = a^n u(n) - a^{n-1} u(n-1)$, find the zero-state response of the system.

Solution: $X(z) = \frac{z}{z-1}, \quad |z| > 1$

$$H(z) = \frac{z}{z-a} - z^{-1} \frac{z}{z-a} = \frac{z-1}{z-a}, \quad |z| > |a|$$

$$Y_{zs}(z) = X(z)H(z) = \frac{z}{\textcolor{red}{z-1}} \cdot \frac{\textcolor{red}{z-1}}{z-a} = \frac{z}{z-a}, \quad |z| > |a|$$

The pole of $X(z)$ cancels with the zero of $H(z)$;
if $|a| < 1$, the ROC of $X(z)H(z)$ will expand.

the zero – state response is: $y_{zs}(n) = a^n u(n)$

Mapping Relationship Between the z-Plane and s-Plane Translation

$$z = e^{sT} \quad T \text{ is the time interval of the sequence}$$

Express s in rectangular coordinate form and **z in polar coordinate form:**

$$s = \sigma + j\omega, \quad z = re^{j\theta}$$

$$z = e^{(\sigma + j\omega)T} = e^{\sigma T} e^{j\omega T} = re^{j\theta}$$

$$\left\{ \begin{array}{l} r = e^{\sigma T} = e^{\frac{2\pi\sigma}{\omega_s}} \\ \theta = \omega T = 2\pi \frac{\omega}{\omega_s} \end{array} \right.$$

$$\omega_s = \frac{2\pi}{T} \text{ is the repetition frequency}$$

7.6 Relationship Between the z-Transform and the Laplace Transform

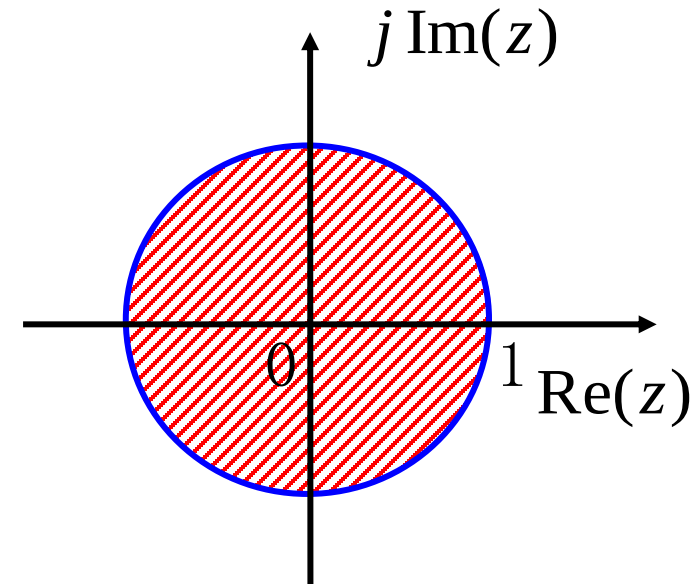
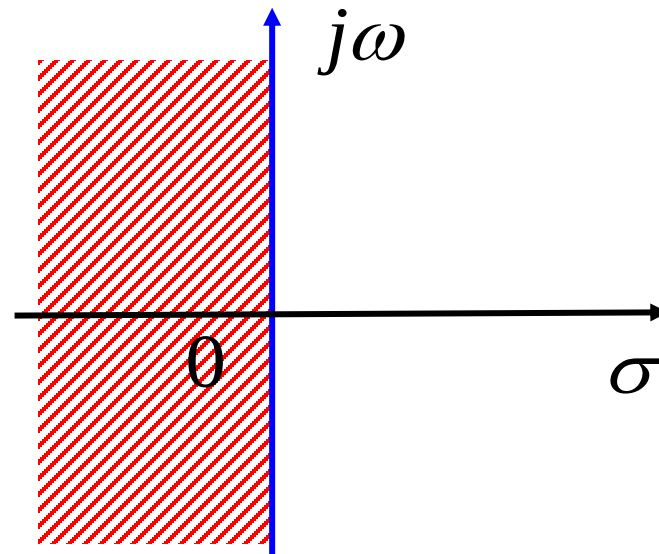


1) The imaginary axis on the s-plane $\xrightarrow{\text{mapping}}$ the unit circle ($r = 1$) in the z-plane

$$\sigma = 0 \quad s = j\omega \quad r = |z| = e^{\sigma T} = 1$$

2) The left half-plane on the s-plane $\xrightarrow{\text{mapping}}$ the interior of the unit circle ($r < 1$) on the z-plane

$$\sigma < 0, s = \sigma + j\omega \quad r = |z| = e^{\sigma T} < 1$$

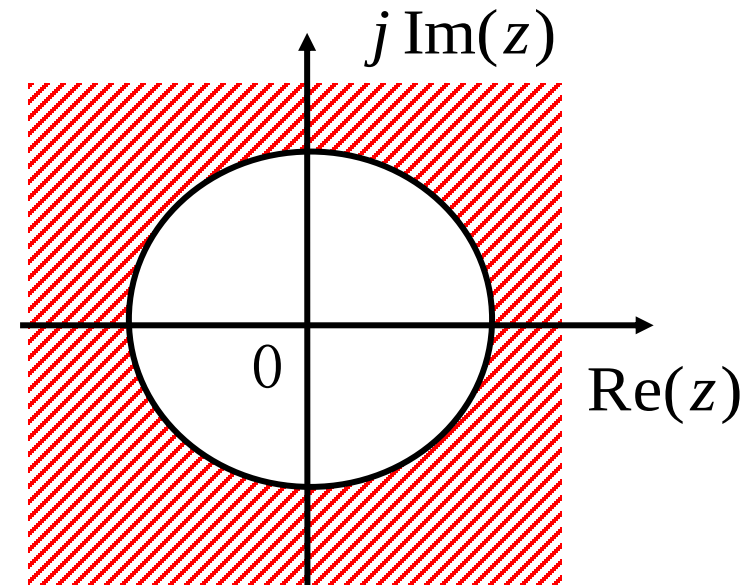
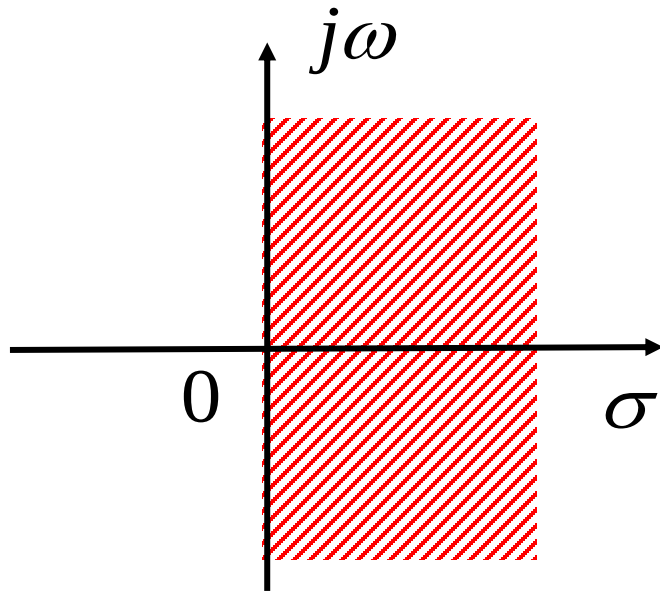


7.6 Relationship Between the z-Transform and the Laplace Transform



3) The right half-plane on the s-plane $\xrightarrow{\text{mapping}}$ the exterior of the unit circle ($r > 1$) on the z-plane

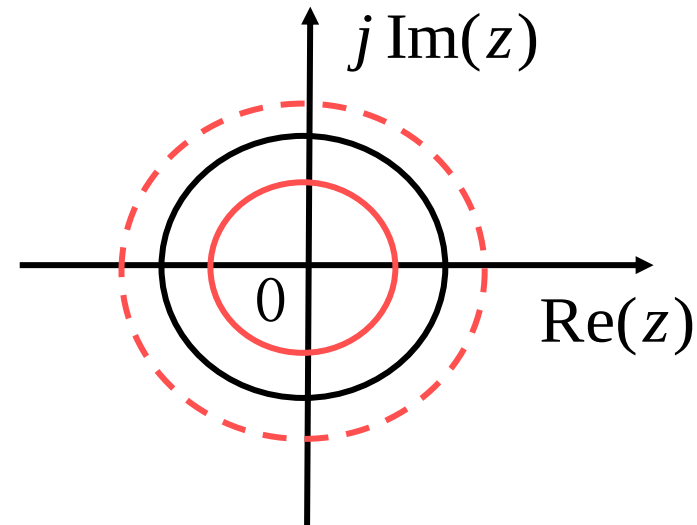
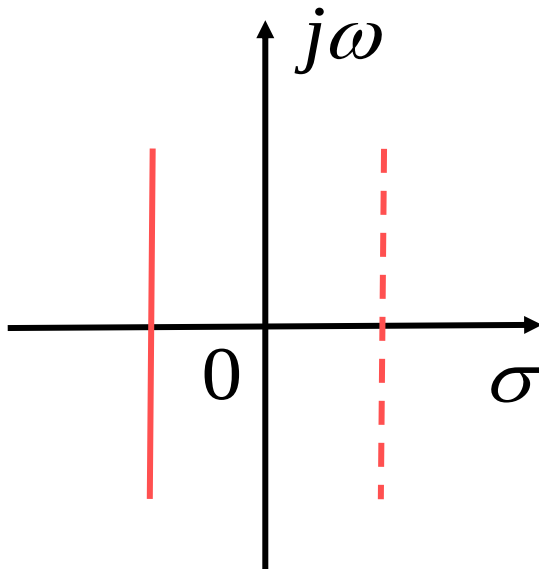
$$\sigma > 0, s = \sigma + j\omega \quad r = |z| = e^{\sigma T} > 1$$



7.6 Relationship Between the z-Transform and the Laplace Transform



4) Lines parallel to the imaginary axis on the s-plane $\xrightarrow{\text{mapping}}$ circles on the z-plane $\begin{pmatrix} \sigma > 0, r > 1 \\ \sigma < 0, r < 1 \end{pmatrix}$



7.6 Relationship Between the z-Transform and the Laplace Transform

5) The real axis ($\omega=0$) on the s-plane $\xrightarrow{\text{mapping}}$ the positive real axis ($\theta=0$) on the z-plane

Lines parallel to the real axis (ω is a constant) $\xrightarrow{\text{mapping}}$

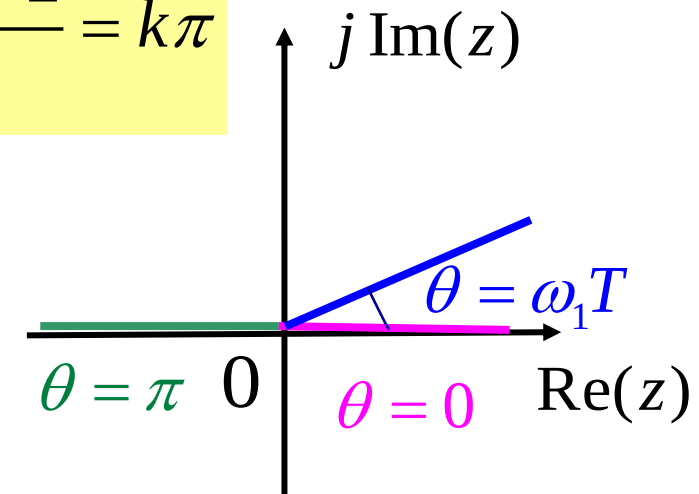
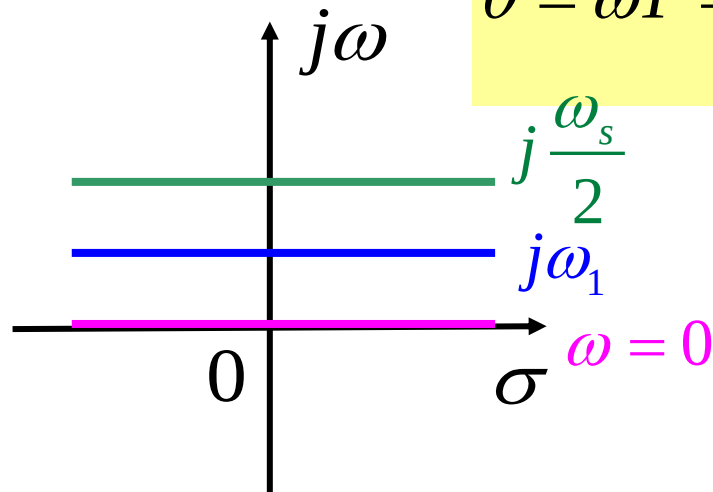
rays originating from the origin on the z-plane

Lines parallel to the real axis (passing through $j \frac{k\omega_s}{2}$ ($k=\pm 1, \pm 3, \dots$)) in the s-plane $\xrightarrow{\text{mapping}}$

the negative real axis in the z-plane $\left(\begin{matrix} \theta = \pi \\ r \text{ is arbitrary} \end{matrix} \right)$

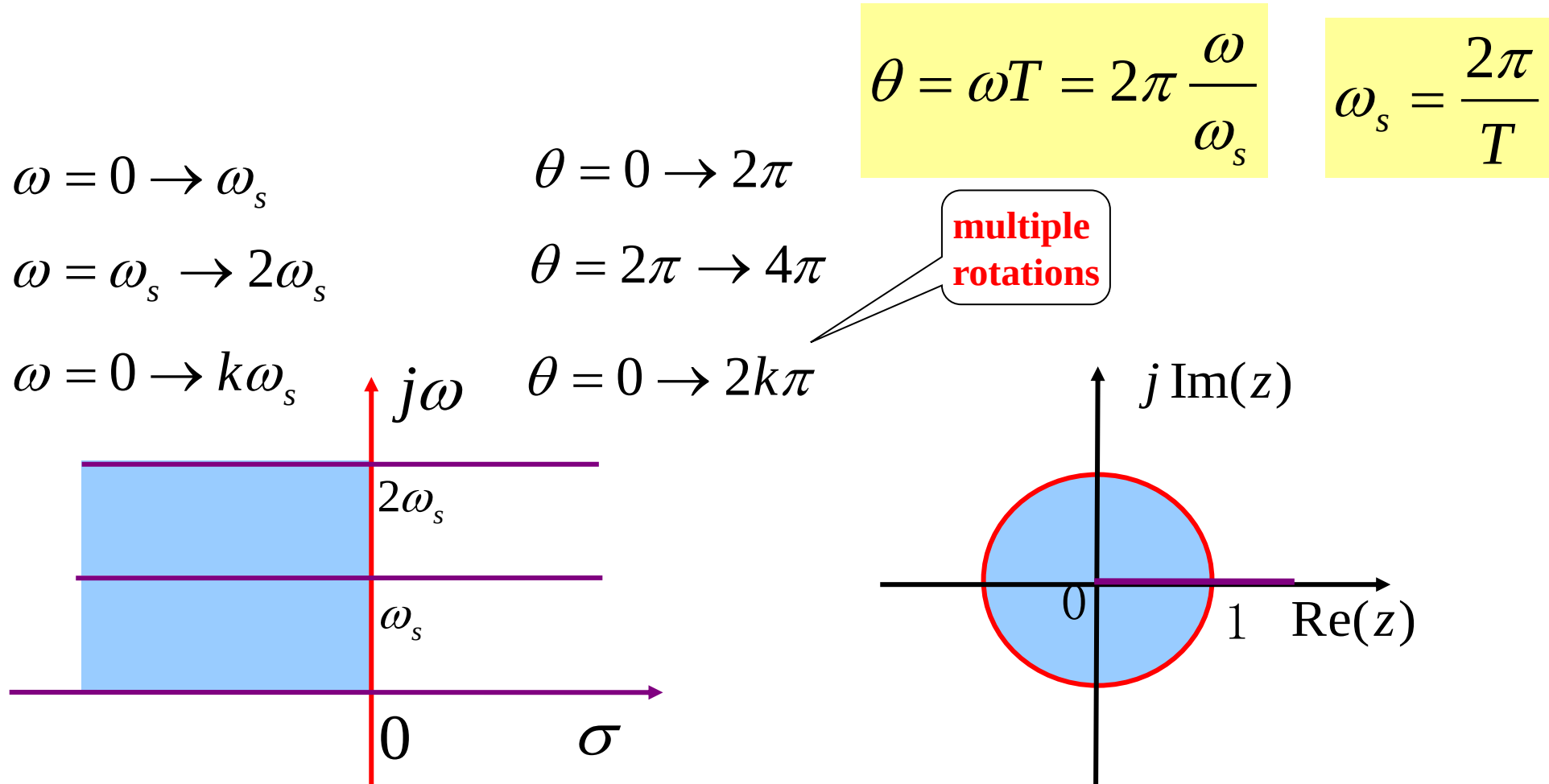
$$\omega_s = \frac{2\pi}{T}$$

$$\theta = \omega T = 2\pi \frac{\omega}{\omega_s} = 2\pi \frac{k\omega_s / 2}{\omega_s} = k\pi$$

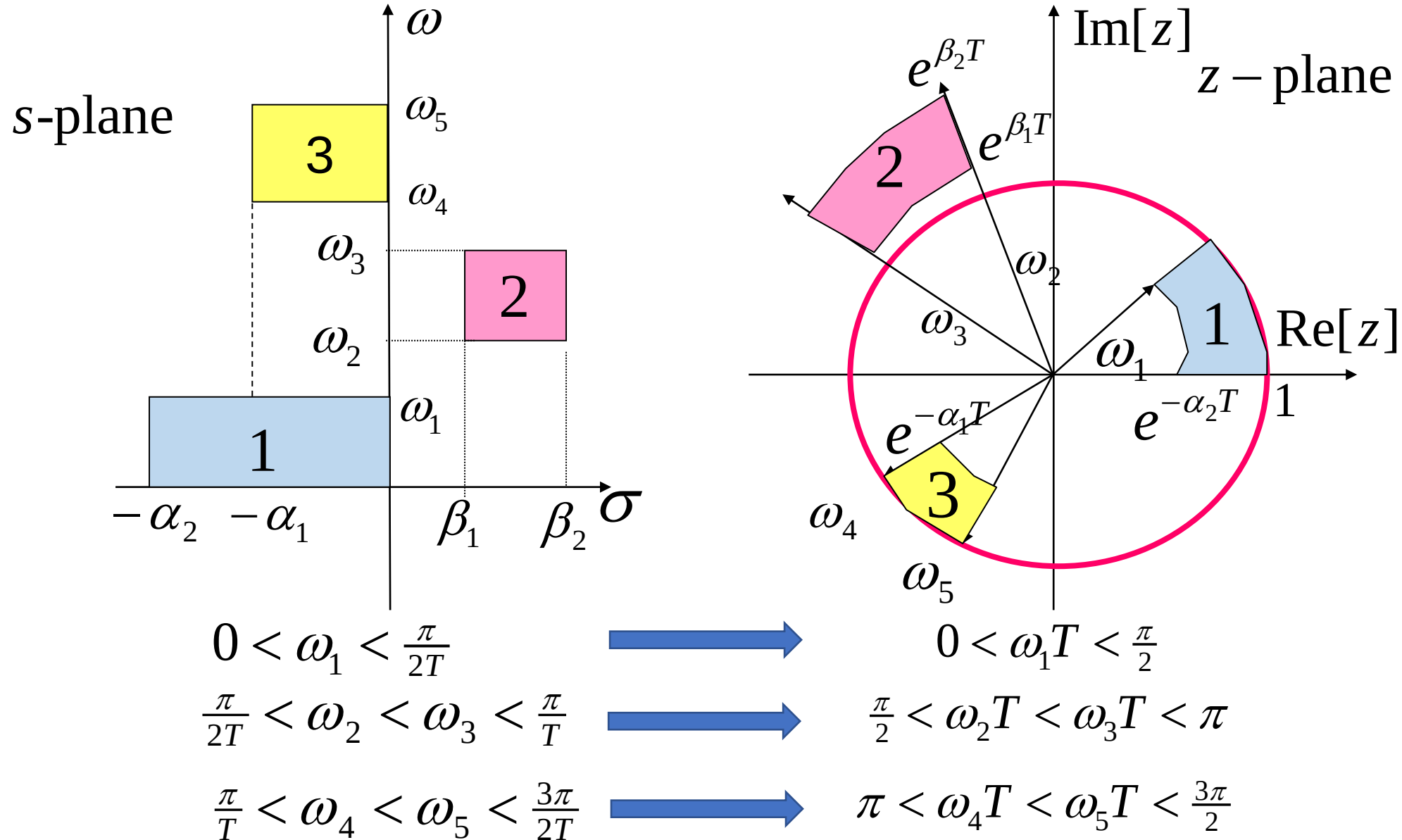


7.6 Relationship Between the z-Transform and the Laplace Transform

6) Moving along the imaginary axis in the s – plane $\xrightarrow{\text{mapping}}$ periodic rotation along the unit circle in the z – plane: each translation by ω_s corresponds to one full rotation around the unit circle, i.e., the mapping between the s-plane and z-plane is not single-valued.



7.6 Relationship Between the z-Transform and the Laplace Transform



1. Repeat all the examples in the lecture notes.
2. 7.21(a)(b)(f) in the textbook.